

VVVC

Vic Voice and Voder Clone
 |
 End-user and VIC-20 wiring guide

Make your VIC speak

VVVC runs a software approximation of the Votrax SC-01 on ESP32 boards. The same speech feeds an always-on GPIO PWM output and, on the original Audio-Kit, its ES8388 codec. It speaks English text or SC-01 phonemes, and can learn phrases in numbered RAM slots for later recall.

One shared speech interface

Send plain text, phonemes or phrase commands through the same connection. There is no mode selector. Board keys have no assigned function; reset restarts the firmware and clears learned phrases.

Start with the board alone

1. On an Audio-Kit, connect a suitable speaker to a labelled speaker output, or use the EARPHONE jack. On any other ESP32, connect the PWM pin through the filter on page 12 to an amplifier. Power the board through USB.
2. Listen for the 880 Hz and 440 Hz startup tones. They check the audio output path without a VIC or terminal.
3. For a speech check, send **-demo** through the USB serial console at 115200 baud. See page 2.

Setting	Current firmware
Hardware	ESP32 / C3 / S2 / S3; Audio-Kit ES8388 is optional
Audio	Always-on 8-bit PWM: GPIO22 (ESP32), GPIO4 (C3/S2/S3); ES8388 is 16-bit
USB console	115200 baud, 8 data bits, no parity, 1 stop bit; no flow control
VIC connection	2400 baud, 8N1; GPIO18 receive, GPIO5 transmit
Learning	80 shared RAM slots; up to 256 SC-01 phonemes per slot

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USB speech and phonemes

Use the Audio-Kit's USB-to-UART connection to send commands from a computer.

Open a terminal

Choose the board's serial port and set **115200 baud, 8N1, no flow control**. Send a CR or LF when you press Enter. Your computer may show the port as COM3, COM4 or another name. Only one terminal or loader can use that port at a time.

With Python installed, run the first line once, then open a terminal. Local echo shows what you type:

```
python -m pip install pyserial==3.5
python -m serial.tools.list_ports
python -m serial.tools.miniterm --dtr 0 --rts 0 -e COM3 115200
```

Press **Ctrl+]** to leave miniterm. Opening a serial port can reset some boards, even with control lines disabled. A reset clears the phrase bank.

Speak plain English

```
HELLO WORLD
She sells seashells on the sea shore
```

Send one line and wait for it to finish. Ordinary text does not return an OK reply. English spelling is converted to SC-01 sounds, so pronunciation is an approximation. The USB limit is 253 bytes per complete line.

Send phonemes directly

```
-phonemes G EH T R EH D Y PA1
-phonemes I0_AH I1_AH I2_AH I3_AH
```

The first example says **Get ready**. Tokens are separated by spaces. PA0 and PA1 are pauses; I0_ through I3_ select inflection. An unknown token rejects the phrase. Successful playback returns **OK PHONEMES**.

Both ports	What happens
-demo	Two tones, HELLO WORLD. THIS IS VIC VOICE., then GET READY; one run
-tone	Repeat the startup tones; reply: Tone complete.
-help	Show commands, audio details, serial settings and phrase-bank limits

Same commands on both ports

On an open VIC channel, use **PRINT#1,"-DEMO"**, **PRINT#1,"-TONE"** or **PRINT#1,"-PHONEMES G EH T R EH D Y PA1"**. Replies return to the sending connection. The demo preserves serial settings and learned slots; it replies **Demo complete.** when finished.

Learn once, recall by number

Slots are shared between USB and the VIC connection.

Choose a slot from **1 to 80**. Learning stores phonemes in RAM without speaking them. Learning into an occupied slot replaces it only after the new phrase passes validation. Use 76-80 for your own phrases when the full Wizard of Wor catalog is loaded.

Command	Purpose
-learn 76 text HELLO WORLD	Convert English now and store the resulting sounds
-learn 77 phonemes G EH T R EH D Y PA1	Store named SC-01 phones, including inflection prefixes
-learn 78 compact ~\{j~?	Store compact codes for G EH T PA1; the backslash is literal
-play 76	Speak the learned phrase; a bare 76 is ordinary text
-slots	List occupied slot numbers and phoneme counts
-forget 76	Delete one slot
-clear	Delete all learned slots

A complete exchange

```
Send:  -learn 76 text HELLO WORLD
Reply: OK 76
```

```
Send:  -play 76
Reply: OK 76
```

The play reply arrives **after speech completes**. Send one command and wait for its reply before sending the next. Replies return on the port that sent the command, with CR/LF line endings.

Limits and error handling

Each slot holds up to 256 phones. Text conversion accepts up to 253 input bytes, but the USB line limit includes the command itself. Named and compact input must also fit the receiving port's line buffer. Compact blocks require an opening marker and final question mark; each data byte must be 0x40-0x7F.

Examples of error replies: **ERROR 76 EMPTY SLOT**, **ERROR 76 INVALID PHONEME**, or **ERROR SLOT MUST BE 1..80**. A rejected replacement leaves the previous phrase intact. -slots reports counts, not the original text.

Temporary means RAM-only

All slots disappear when the ESP32 resets or loses power. They are not written to flash. Resetting only the VIC leaves a separately powered ESP32's bank intact. Normal speech and baud changes preserve it.

Load Wizard of Wor

The supplied host script teaches the original catalog through normal learning commands.

The firmware starts with empty slots. **tools/learn_wizard.py** loads all 75 catalog entries into slots 1-75, preserving the original numbering, phonemes and inflection. Slots 76-80 are left alone. The complete catalog is included with VVVC; no separate phrase source is needed.

Load the whole catalog

Close the serial terminal first. In the VVVC project folder, run:

```
python -m pip install -r tools/requirements.txt
python tools/learn_wizard.py --port COM3
```

The script prints each slot as it receives OK, then checks the returned phoneme count for every loaded slot. A successful run ends with **Verified 75 phrases**. It replaces any existing content in slots 1-75.

Play immediately, or load a selection

```
python tools/learn_wizard.py --port COM3 --play 26
python tools/learn_wizard.py --port COM3 --phrases 21 26 55
python tools/learn_wizard.py --list
python tools/learn_wizard.py --dry-run
```

--play loads and verifies first, then speaks the requested slot once. --phrases loads only the listed original IDs. --list shows the catalog; --dry-run prints the learn commands without connecting.

Recall	Phrase
-play 21	Get ready, warrior.
-play 24	Hey, insert coin.
-play 26	I am The Wizord of Wor.
-play 46	Remember, I'm the wizard, not you!
-play 55	Welcome, to my world of wor.

These are the original catalog labels, including their spelling. The full number list is on pages 13-14. Recall works from either connection; there is no separate Wizard mode or background random playback.

Keep the board powered

Reload after an ESP32 reset. Some terminal programs reset the board when they connect; --play lets you test without reopening a terminal. Older VVVC firmware with fewer than 75 slots cannot hold the full catalog.

Speak from VIC BASIC

Complete the wiring checks on pages 6-9 before connecting the computer.

Device **2** is the VIC's user-port serial channel. `CHR$(10)` selects **2400 baud, 8N1**, matching the ESP32 default. OPEN before assigning variables: the VIC clears them to allocate its serial buffers. Allow at least 512 bytes of free BASIC memory. [2]

First words

```
10 OPEN1,2,3,CHR$(10)
20 PRINT#1,"HELLO WORLD"
30 FOR I=1 TO 1000:NEXT
40 CLOSE1
```

The delay gives the VIC's transmit buffer time to empty before CLOSE. For interactive testing, enter OPEN and PRINT as direct commands, wait to hear the phrase, then enter CLOSE1. Do not keep sending while speech is still playing.

Recall a phrase learned from either port

```
10 OPEN1,2,3,CHR$(10)
20 PRINT#1,"-PLAY 26"
30 FOR I=1 TO 1000:NEXT
40 CLOSE1
```

This says the learned Wizard phrase in slot 26, provided the loader has run since the last ESP32 reset. It works even if PSEND is on.

Teach slots directly from the VIC

```
OPEN1,2,3,CHR$(10)
PRINT#1,"-LEARN 76 PHONEMES G EH T R EH D Y PA1"
PRINT#1,"-PLAY 76"
CLOSE1
```

Enter individually: wait for OK 76 before PLAY, then wait for speech before CLOSE. Page 9 reads replies automatically. For immediate speech without learning, use **PRINT#1,"-PHONEMES G EH T R EH D Y PA1"**.

Send compact SC-01 bytes directly

```
10 OPEN1,2,3,CHR$(10)
20 PRINT#1,CHR$(222);CHR$(92);CHR$(123);
30 PRINT#1,CHR$(106);CHR$(126);"? "
40 FOR I=1 TO 1000:NEXT
50 CLOSE1
```

`CHR$(222)` opens the block; bytes 92, 123, 106 and 126 encode G, EH, T and PA1. The ? closes it. Line 20's final semicolon joins the PRINT statements. `CHR$` avoids PETSCII keyboard ambiguities.

Identify the right contacts

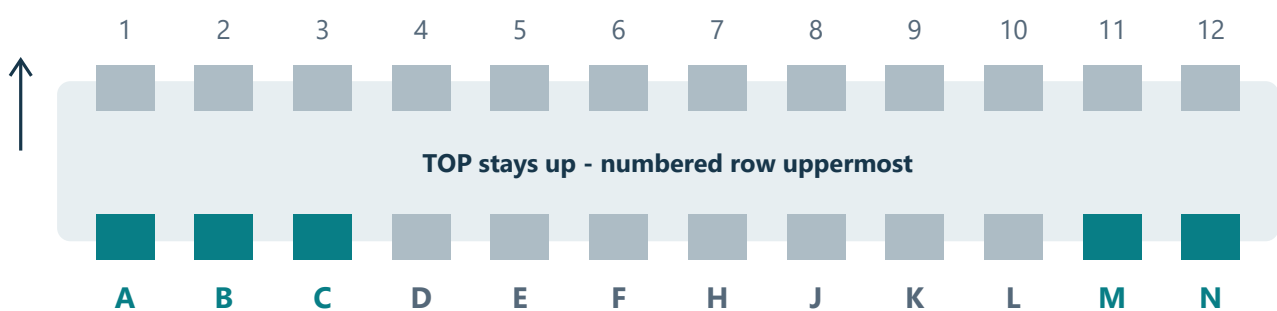
The VIC has a user-port edge connector; the GPIO numbers belong to the ESP32.

Power off before fitting or changing the cable

Disconnect both power sources while wiring. This guide is for the **VIC-20 user port**, not the round IEC socket, cartridge connector or a PC RS-232 connector. Leave the VIC's supply pins disconnected; power the Audio-Kit separately by USB.

VIEW A: LOOKING INTO THE VIC FROM BEHIND

VIC upright; you face its rear, looking toward the keyboard.



This is the computer's port, viewed from outside at the rear. The VIC sits normally with its keyboard uppermost. Contact 1 is upper left; A is lower left, and N is lower right. The letter sequence omits G and I. The matching cable views are on page 7. [1]

VIC contact	VIC-20 signal	Connect to
M	CB2 - transmitted data	GPIO18 RX through the 10 kOhm / 18 kOhm divider
B	CB1 - receive interrupt	GPIO5 TX return branch
C	PB0 - received data	The same GPIO5 TX signal as B
A or N	Ground	ESP32 GND and divider ground

Parts for the minimal interface

A correctly keyed 24-contact (2 x 12) VIC user-port connector; short insulated wire; one 10 kOhm resistor and one 18 kOhm resistor (1%, 0.25 W are suitable); optionally two 470 Ohm series resistors for the return branches; insulation and strain relief; and a multimeter.

Read the board labels

Use pads or header pins marked GPIO18/IO18, GPIO5/IO5 and GND. Header position varies by board revision. These are not physical pin numbers 18 and 5 on the module. Leave Audio-Kit keys 5 and 6 unused because they share GPIO18 and GPIO5.

Cable views: avoid mirroring

These drawings show the same plug from opposite ends. Its installed TOP stays up.

Mark TOP before soldering

Mark the side of the plug that will face the top of the upright VIC. When changing viewing ends, keep this mark uppermost: turn the plug around horizontally, not upside down. The numbered row stays above the lettered row in both views.

VIEW B: LOOKING INTO THE CABLE'S MATING OPENING

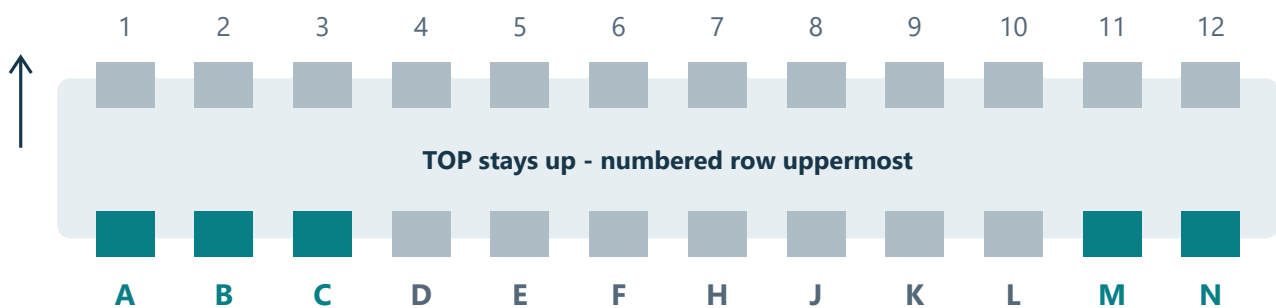
Cable unplugged; look into the slot, toward the cable wires.



Mating face: this is the opening that slides onto the VIC's edge contacts. With TOP up, contact 12 is upper left and N is lower left. This view is the left/right mirror of the computer's rear-port view.

VIEW C: LOOKING AT THE CABLE'S WIRE / SOLDER SIDE

Look past the wire terminals toward the VIC; the mating opening faces away.



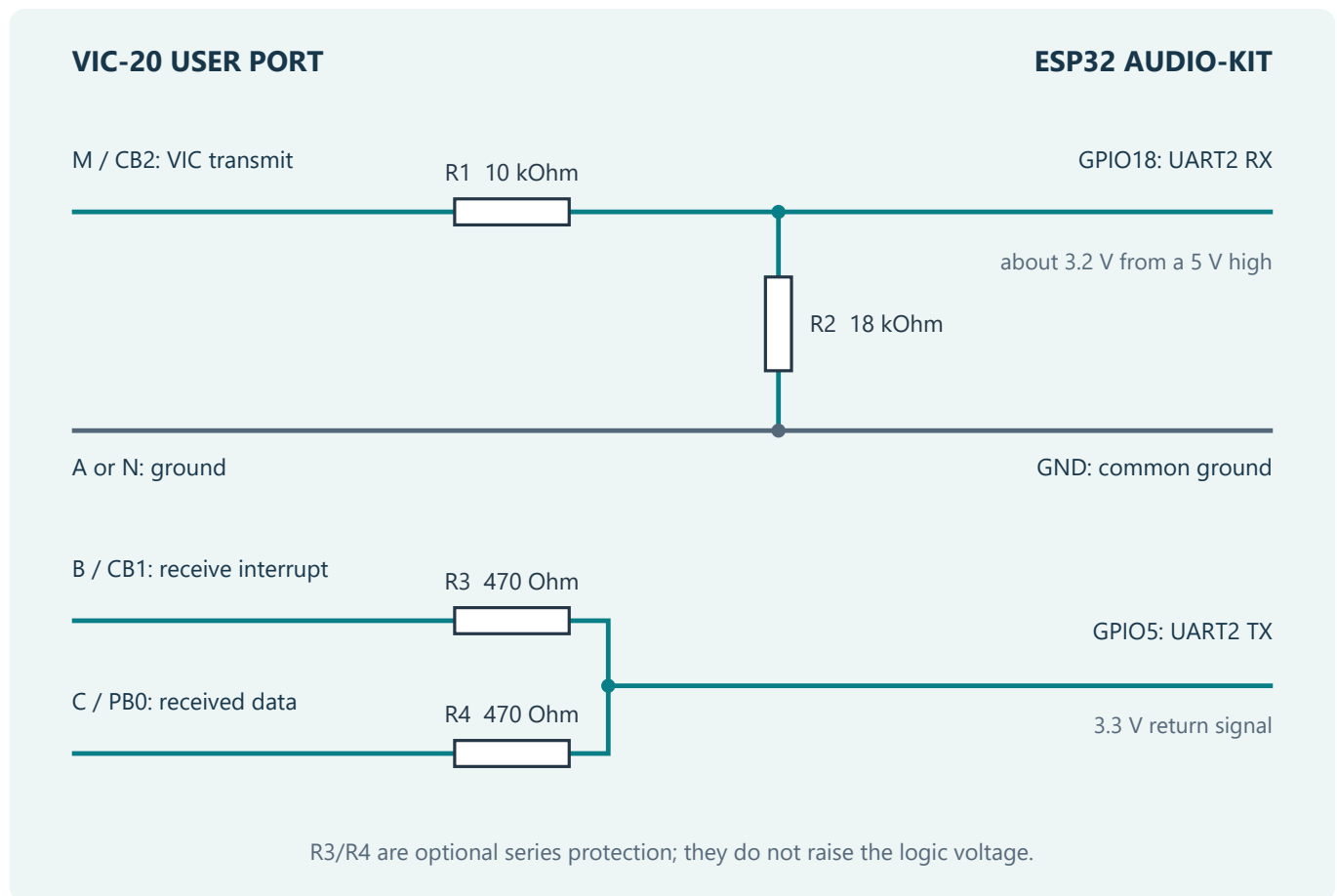
Wire side: you stand behind the cable, looking toward the VIC, as when the plug is installed. Contact 1 is upper left and A is lower left. This ordering matches View A on page 6; it is the mirror of View B above.

Verify contacts, not just the drawing

The maps identify electrical contacts; a particular connector may stagger or rearrange its solder tails. Use moulded pin markings, its datasheet, and a continuity meter to map each wire terminal to the mating contact. Check M, B, C and A/N with both devices unpowered.

Wire the serial interface

Logical connection diagram: trace each named signal, rather than copying physical positions.



Protect the ESP32 receive input

Put R1 (10 kOhm) **between VIC M and GPIO18**. Put R2 (18 kOhm) **from the GPIO18 junction to ground**. Do not swap them. With a 5 V input, the divider gives $5 \times 18 / (10 + 18) = 3.21 \text{ V}$. ESP32 GPIO is 3.3 V logic; do not feed the VIC's 5 V signal straight into it. [3]

Return data needs both B and C

GPIO5 feeds the VIC's data input C and interrupt input B. The optional 470 Ohm resistors limit contention current; they are not level shifters. This minimal circuit assumes the VIC's fitted VIA accepts a 3.3 V logic high. Verify this when a VIA has been replaced or return data is unreliable.

Check actual levels before the first run

With the divider junction still disconnected from GPIO18, check the idle high after opening the VIC channel. Aim for about 3.0-3.3 V. At 3.3 V supply, the ESP32 high-input minimum is about 2.48 V [3]. If it is too low, too high or negative, stop and use a correctly specified non-inverting logic-level interface; do not connect it as-is.

Keep signal leads short and connect grounds. No MAX232 or RS-232 voltage adapter belongs in this direct logic-level circuit. The speaker stays on the Audio-Kit's speaker terminals; GPIO25 is the codec word clock, not an audio output for an external speaker.

Bring up the VIC connection

Test one direction at a time, then check replies before using long transfers.

- 1. With power off:** check pin letters, divider values, common ground, insulation, and no shorts to the VIC supply or reset contacts.
- 2. Power the Audio-Kit:** confirm startup tones, as in the board-only test. Power both devices for serial use; avoid leaving signal wires driven into an unpowered device.
- 3. Check VIC transmit:** after the voltage check on page 8, power off, finish the M-to-GPIO18 connection, then try HELLO WORLD from page 5.
- 4. Check the return:** with power off, add GPIO5 to both B and C. Run the program below. It disables echo, learns slot 76, waits for OK, recalls it, and waits for the second OK.

```

10 OPEN1,2,3,CHR$(10)
20 PRINT#1,CHR$(27);CHR$(20)
30 PRINT#1,"-LEARN 76 TEXT HELLO WORLD"
40 GOSUB200
50 IF LEFT$(R$,2)<>"OK" THEN90
60 PRINT#1,"-PLAY 76"
70 GOSUB200
90 CLOSE1:END
200 R$="":T=TI
210 GET#1,A$
220 IF A$="" THEN IF TI-T<1200 THEN210
230 IF A$="" THEN R$="TIMEOUT":PRINT R$:RETURN
240 IF A$=CHR$(10) OR A$=CHR$(27) THEN210
250 IF A$=CHR$(13) THEN PRINT R$:RETURN
260 R$=R$+A$:GOTO210

```

Expect two **OK 76** replies with HELLO WORLD between them. The routine ignores LF and the initial echoed ESC, ends on CR, and times out after about 20 seconds. GET# allows a timeout; INPUT# could wait indefinitely. [2]

If a test fails

Speech works but replies time out: check both B and C, the return signal level, and that GPIO5 is not held low by key 6.

No VIC speech, but USB works: recheck M, GPIO18, the divider, ground and 2400-baud settings. GPIO18 is shared with key 5.

Intermittent results: shorten leads and check logic levels. A properly specified non-inverting 5 V/3.3 V serial level interface is preferable to guessing resistor changes.

VIC controls and baud rates

Escape controls are VIC-port bytes, not USB terminal commands.

Bytes	BASIC value after CHR\$(27)	Action
ESC, 0x11	CHR\$(17)	PSEND on: return compact phonemes instead of speaking text
ESC, 0x12	CHR\$(18)	PSEND off: normal speech
ESC, 0x13	CHR\$(19)	Echo on (startup default)
ESC, 0x14	CHR\$(20)	Echo off; recommended when reading command replies
ESC, 0x15	CHR\$(21)	CAPS spelling on
ESC, 0x16	CHR\$(22)	CAPS spelling off (startup default)
ESC, 0x17	CHR\$(23)	Disable the four-second inactivity timer until reset
ESC, 0x18	CHR\$(24)	Restart the ESP32; clears learned slots

For example, on an already open channel: **PRINT#1,CHR\$(27);CHR\$(20)**. Learning replies are returned even with echo off. PSEND and CAPS do not change how text is learned; explicit -play still speaks.

Change from 2400 to 1200 baud

```
OPEN1,2,3,CHR$(10)
PRINT#1,"SET TTY LO"
CLOSE1
OPEN1,2,3,CHR$(8)
```

Enter these as separate direct commands, allowing transmission to complete before CLOSE. Send the change at the **current** speed, then reopen at the new speed. CHR\$(8) selects 1200; CHR\$(10) selects 2400. To return, send SET TTY HI at 1200, close, then reopen with CHR\$(10). Baud changes preserve the phrase bank; an ESP32 reset restores 2400 and clears it.

Framing and flow

Press RETURN to end a line. There is no hardware flow control, so wait for replies and let speech finish before sending the next request. Commands must end with RETURN. For compact phoneme blocks, use CHR\$ for exact byte values.

Troubleshooting

Use the simplest working test to narrow down a problem.

Symptom	Check
No startup tones	Check USB power and the speaker or EARPHONE connection. Use a supported ES8388 Audio-Kit for codec audio.
Tones work, speech does not	Try -demo over USB. For ordinary VIC text, turn PSEND off. Use valid named phones or a previously learned slot.
ERROR n EMPTY SLOT	Reload after reset. Wizard data is supplied by the host script; it is not built into the firmware.
Cannot open COM port	Close other terminals or flashing tools. Use a USB data cable and the board's USB-to-UART port.
Learn failed / line too long	Keep the whole USB line within 253 bytes. A slot allows 256 phones. Check the format word and wait for each reply.
VIC fails, USB works	Use pages 6-9: ground, divider, M/B/C, RX/TX directions, voltage levels, and matching baud rates.
Scott Adams on Mega-Cart	After loading the game from the Mega-Cart menu, hold F1 while pressing the Mega-Cart reset button. Then enter SYS32592 and press RETURN. This initializes the VIC's KERNAL serial routines; without it, speech output can be corrupted or unintelligible.
VIC help/list output is incomplete	Read -help and -slots replies as they arrive. Long output can overrun the VIC receive buffer if BASIC cannot keep up; USB is useful for inspection.

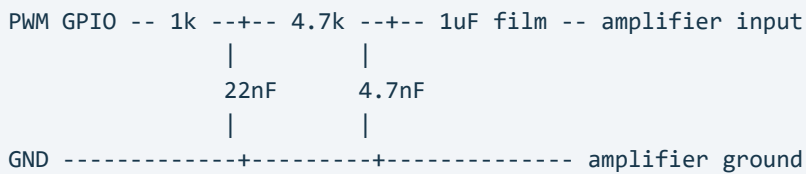
GPIO audio output

Use this connection with ESP32 boards that do not have the Audio-Kit audio codec.

The Audio-Kit's speaker and headphone outputs need no extra filter. On other supported boards, the speech signal is also available as PWM on GPIO22 for the original ESP32, or GPIO4 for ESP32-C3, S2 and S3. Connect it to an amplifier through this filter. Never connect a speaker directly to a GPIO pin.

Connect to an amplifier

Use an amplifier input impedance of 47 kOhm or greater. Start with low volume. The capacitors reduce the PWM carrier; the 1 uF film capacitor blocks DC:



[1] Commodore, **VIC-20 Personal Computing Guide**, printed p. 152, user-port contacts. [Read the scanned manual.](#)

[2] Commodore, **VIC-20 Programmer's Reference Guide**, pp. 176 and 251-256; pin functions, serial buffers, OPEN and GET#. [Read the transcribed manual.](#)

[3] Espressif, **ESP32 Series Datasheet**, DC input characteristics. [Read the manufacturer datasheet.](#)

Board family: [Ai-Thinker Audio-Kit documentation](#).

Wizard of Wor: 1-38

Run the loader first, then recall with -play N. Original catalog wording is retained.

Slot	Phrase label
1	A few more dungeons and you'll be a worlord.
2	Ah good, my pets where getting hungry. ha ha ha ha.
3	Ah you thought you could but I'm the dungeon master. Ha ha ha ha.
4	And my teleporting spell can be even faster. Ha ha ha ha.
5	Another coin for my treasure chest.
6	Another worrior for my babies to devour.
7	Are you fit to survive the pit. Ha ha ha ha.
8	Be forewarned, you approach the pit. Ha ha ha ha.
9	Bite the bullet worlord. Ha ha ha ha.
10	Bite the bullet worrior. Ha ha ha ha.
11	Burwor hasn't eatin anyone in months. ha ha ha ha.
12	Burwor, Garwor, and Thorwor, will do you in.
13	Burwor, you are in the worlord dungeons.
14	Come back for more, with, The Wizard of Wor. ha ha ha ha.
15	Deep in the caverns of wor you will meet me worrior.
16	Deeper ever deeper in to the dungeons of wor. Ha ha ha ha.
17	Fear, I draw near, each time I appear. ha ha ha ha.
18	Find me, The Wizard of Wor.
19	Garwor and Thorwor become invisible. ha ha ha ha.
20	Garwor, go after them.
21	Get ready, worrior.
22	Hay, Your space boots untied. Ha ha ha ha.
23	He he he ho ho ho ha ha ha ha, that was fun.
24	Hey, insert coin.
25	Hurry back, I can't wait to do it again.
26	I am The Wized of Wor.
27	I'll fray you with my lightning bolts.
28	I'm out'a spite. ha ha ha ha.
29	If you can't beat the rest, then you'll never get the best. Ha ha ha ha.
30	If you destroy my babies, I'll pop you in the oven. ha ha ha ha.
31	If you get too powerful, I'll take care of you myself. ha ha ha ha.
32	If you try any harder, you'll only meet with doom. ha ha ha ha.
33	Keep going, and you will find me.
34	Kill Worluk for double score.
35	My babies breath fire, worrior.
36	My beasts run wild in the worlord dungeons. Ha ha ha ha.
37	My creatures are radioactive.
38	My magic is stronger then your weapons, worlord.

Wizard of Wor: 39-75

Run the loader first, then recall with -play N. Original catalog wording is retained.

Slot	Phrase label
39	My magic is stronger then your weapons, warrior.
40	My worlings are very very hungry. Ha ha ha ha.
41	Now you get the heavyweights.
42	Now you know the tast of my magic, warrior.
43	Now your only chance is your dance. Ha ha ha ha.
44	One bite from my pretties, and you'll explode. Ha ha ha ha.
45	Oops, I must have forgotten the walls. Ha ha ha ha.
46	Remember, I'm the wizard, not you!
47	Say it again, Woryer fear, I draw neer, each time I apear. Ha ha ha ha.
48	So you've come to score in the world of wor. Ha ha ha ha.
49	The dungeons of wor await your return, warrior.
50	The Wizard of Wor thanks you.
51	Thor, Bur, Gar dinner's ready. Ha ha ha ha.
52	Thorwor is red mean and hungry for spice food.
53	Wasn't that lightning bolt delicious? Ha ha ha ha.
54	Watch the radar, warrior.
55	Welcome, to my world of wor.
56	Were are you going to hide now. Ha ha ha ha.
57	Worlock will escape through the door.
58	Worlord maybe you'll see me again.
59	Worlord, while you developed science, we developed magic.
60	Worlord, You won't have a chance for your dance. Ha ha ha ha.
61	Worlord, Your path leads directly to the pit. Ha ha ha ha.
62	Worrior, now I'm getting mad.
63	Worrior, while you developed science, we developed magic.
64	You are in, the dungeons of wor.
65	You can start anew, but for now you're through.
66	You know you can do better, worlord.
67	You know you can do better, warrior.
68	You will never leave wor alive. ha ha ha ha.
69	You'd better hope you don't find me, The Wizard of Wor.
70	You'll get the arena. Ha ha ha ha.
71	You're asking for trouble, warrior.
72	You're off to see the wizord, the magical Wizord of Wor. ha ha ha ha.
73	You've just been frayed by the Wizard of Wor. Ha ha ha ha.
74	Your bones will lie in the dungeons of wor. ha ha ha ha.
75	Your explosion was music to my ears. Ha ha ha ha.

Slots 76-80 are spare. Resetting or powering off the ESP32 clears all slots.